

Medical Cost Prediction Using Machine Learning

GTU Data Science Mini Project | Industry-Oriented Healthcare Analytics
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PROBLEM STATEMENT

Healthcare costs are rising globally.
Insurance firms need accurate premium estimates to reduce financial risk.

Goal: Predict annual insurance charges (USD) from age, BMI, smoking, region, and family details using ML.

METHODOLOGY

1. Collect Medical Cost dataset (1338 rows)
2. Clean data; encode categories (One-Hot)
3. Train/Test split 80/20
4. EDA: age, BMI, smoker vs charges
5. Random Forest Regressor (350 trees)
6. Evaluate: MAE, RMSE, R²
7. Deploy: notebook + Streamlit web app

TOOLS USED

- Python 3
- pandas & NumPy — data handling
- scikit-learn — ML pipeline
- matplotlib & seaborn — charts
- Jupyter Notebook — analysis
- Streamlit — live prediction UI
- GitHub — project portfolio

OUTPUT & RESULTS

\$1,982

MAE

\$2,517

RMSE

0.93

R² Score

Sample Prediction: Female, 39 yrs | BMI 29.4 | 2 children | Non-smoker | Northeast

Predicted Annual Charge: \$14,623

Deliverables: Jupyter notebook • insurance.csv dataset • bulk_predictions CSV • Streamlit app

Societal Impact: Fair insurance pricing | Hospital planning | Patient cost awareness

Pipeline: Data → Preprocess → Visualize → Train Model → Evaluate → Predict